

DATA VISUALIZATION WITH MICROSOFT POWER BI

Comparative Study of Stock Market Data

Apple • Google • Tesla

Purpose

The purpose of this data visualization task is to present cleaned stock-market data clearly in Microsoft Power BI and carry out a comparison. The dashboards use line charts, bar charts, scatter plots, and date filters to examine stock prices, trading volume, and connections between variables. The analysis aims to identify important trends, patterns, relationships, and differences among Apple, Google, and Tesla.

Important Stock Variables

Open refers to the stock price at the start of the trading period. High represents the maximum price reached during the session, whereas Low represents the minimum price. Close is the final price recorded for the day. Adjusted Close considers corporate actions such as stock splits and dividends, making it useful for historical comparison. Volume indicates the number of shares traded during the period, with higher values generally reflecting increased trading activity.

Visualizations Applied

Visualization	Purpose	Variables / Measurement
Line chart	Track stock-price changes and trends over time	Year with Total Close / Total Open
Bar chart	Compare trading activity between years	Year with Total Volume
Scatter plot	Study relationships between numerical values	Open vs Close / High vs Low
Date slicer	View the dashboard for a chosen time period	Date
Count distribution chart	Examine how records are distributed across price values	Date count by Open / Close

Analysis of Apple Stock

The image below shows the Power BI dashboard provided for this company and serves as the supporting evidence chart for the analysis.

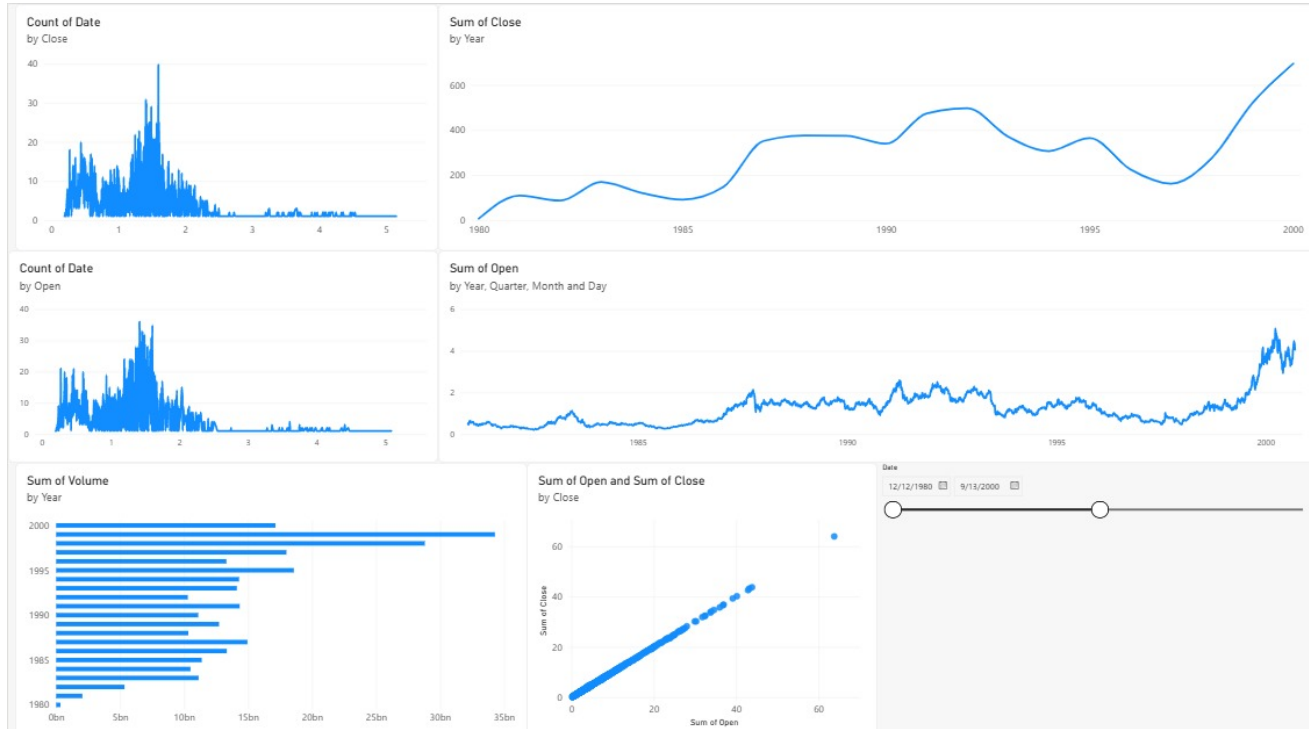


Figure: Power BI dashboard evidence chart for Apple Stock Analysis

Trends Identified from the Graphs

The closing-price line generally moves upward over the displayed historical period, although it includes several temporary increases and decreases.

The opening-price series broadly follows the same direction as the closing-price series, with some short-term variation.

The yearly volume chart indicates that trading activity differs from one year to another, with certain years showing considerably higher volume.

The Open-versus-Close scatter plot indicates a strong positive relationship, where higher opening prices are generally linked with higher closing prices.

The date filter makes it possible to study a particular period in greater detail.

Comparison Observations

The Apple dashboard shows a mostly positive long-term price direction, although the movement is not continuously upward.

Trading volume differs across years, indicating changes in market participation.

The strong Open-Close relationship suggests that the two daily price measures tend to move closely together.

Overall Analysis

The Apple dashboard brings together price trends, trading activity, and relationships between prices. The line chart helps show long-term movement, the volume chart highlights changes in activity, and the scatter plot illustrates the close connection between opening and closing prices.

Analysis of Google Stock

The image below shows the Power BI dashboard provided for this company and serves as the supporting evidence chart for the analysis.

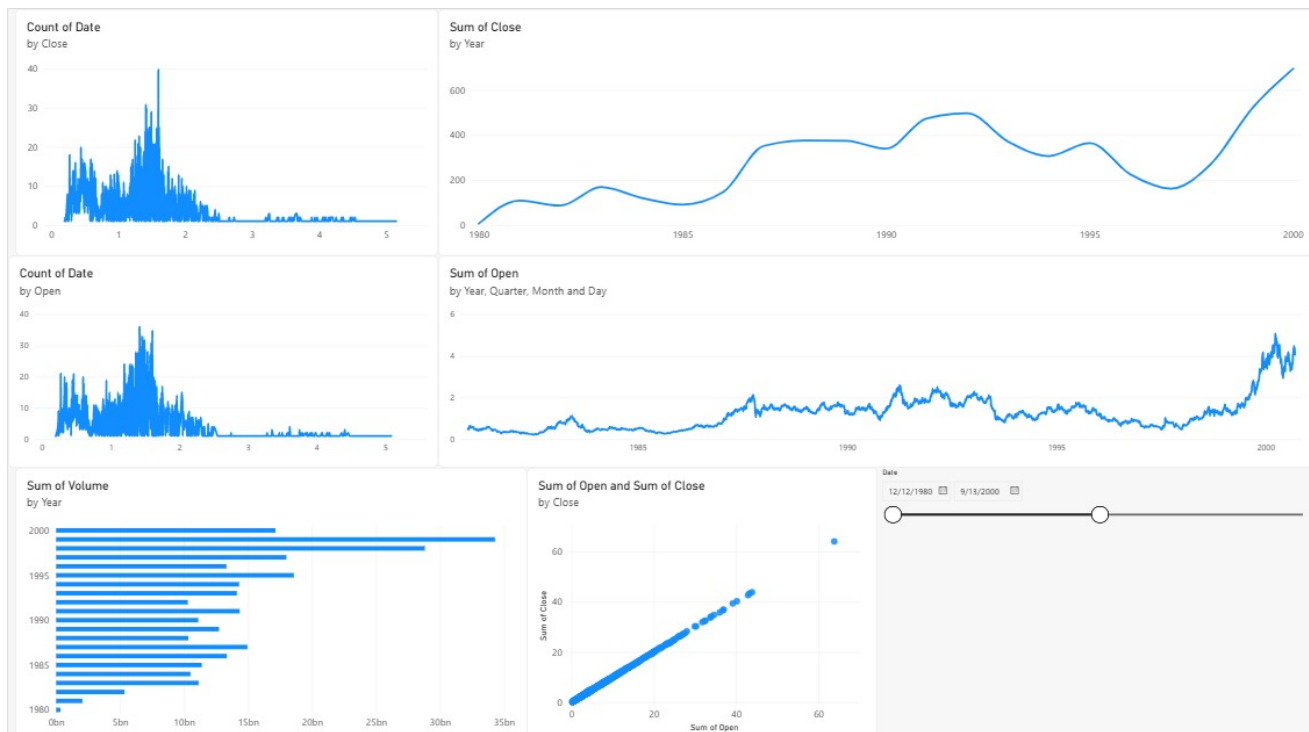


Figure: Power BI dashboard evidence chart for Google Stock Analysis

Trends Identified from the Graphs

The closing-price line shows a general upward direction throughout the displayed period, with some temporary slowdowns and decreases.

The opening-price line also has an overall upward direction, while showing several short-term fluctuations.

The annual volume chart shows that trading activity changes over time, with some years recording much higher volume than others.

The High-versus-Low scatter plot displays a strong positive relationship, meaning higher high-price values are generally linked with higher low-price values.

The date slicer allows the analysis to be narrowed to a chosen period.

Comparison Observations

Google shows overall long-term growth along with short-term market variations.

Changes in yearly volume reflect different levels of market participation.

The strong High-Low relationship shows that the two price measures generally move together.

Overall Analysis

The Google dashboard presents an overall positive price direction while also showing normal short-term variation. Together, the line, volume, scatter, and date-filter visuals provide a clear picture of long-term movement and changes in trading activity.

Analysis of Tesla Stock

The image below shows the Power BI dashboard provided for this company and serves as the supporting evidence chart for the analysis.

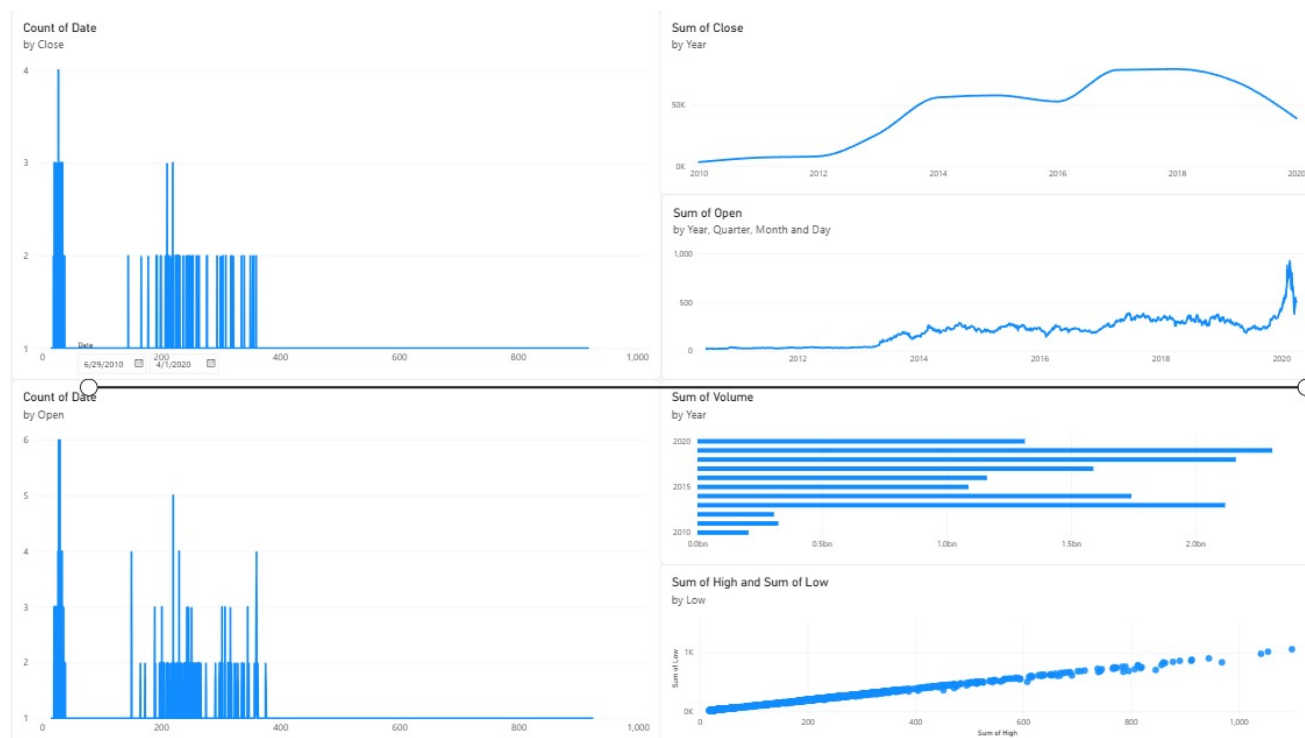


Figure: Power BI dashboard evidence chart for Tesla Stock Analysis

Trends Identified from the Graphs

The closing-price line rises sharply during the earlier part of the displayed period, reaches a high

point, and then declines toward the final year shown.

The opening-price series records strong growth and noticeable volatility, including a significant spike near the end of the displayed period.

The annual volume chart shows changing trading activity, with several later years recording much higher volume than some earlier years.

The High-versus-Low scatter plot shows a clear positive relationship, where higher high-price values are generally accompanied by higher low-price values.

The date slicer can be used to select a particular time period for more detailed examination.

Comparison Observations

Tesla displays a stronger rise-and-fall pattern in the supplied dashboard.

The increase in trading volume during later years suggests higher market activity in those periods.

The strong High-Low relationship remains noticeable even with Tesla's larger price fluctuations.

Overall Analysis

The Tesla dashboard highlights substantial growth followed by a noticeable decline and higher price volatility. The combination of price, volume, relationship, and date-filter charts gives a useful overview of Tesla's changing market behavior.

Comparison: Apple vs Google vs Tesla

The three dashboards can be compared based on price movement, trading activity, and relationships among numerical variables. Each dashboard uses time-based line charts to show trends and bar charts to compare yearly trading volume. Scatter plots add further insight into relationships between price measures.

Comparison Aspect	Apple	Google	Tesla
General price direction	Mostly upward with variations	Generally upward with brief slowdowns	Sharp rise followed by a noticeable decline near the end
Price variation	Noticeable variations	Moderate variations	Stronger variations
Trading activity	Changes from year to year	Changes from year to year	Later years show clearly higher activity

Relationship between
prices

Open and Close closely
related

High and Low closely
related

High and Low closely
related

Main Findings

Line charts are useful for recognizing long-term stock-price trends as well as short-term fluctuations.

Bar charts demonstrate that yearly trading volume varies over time.

Scatter plots highlight strong positive relationships between related stock-price measures.

All three companies experience periods of growth and fluctuation, while Tesla's dashboard presents the most noticeable rise-and-fall pattern.

Date slicers add interactivity and allow users to examine selected periods.

Combining multiple visualizations provides a broader understanding of price movement, market activity, and relationships.

Final Conclusion

The Power BI dashboards effectively present and compare historical stock-market data for Apple, Google, and Tesla. The visualizations show long-term price trends, variations in yearly trading activity, and relationships among numerical price measures. Interactive date filters make it easier to examine specific periods in detail. Overall, the dashboards demonstrate how cleaned stock-market data can be transformed through visualization into useful patterns, trends, relationships, and comparisons.